

AI-Based Billboard Size Estimation Using Computer Vision

A Technical Research Report | Author: Anuradha Pandit | Year: 2026

Abstract

This report presents an AI-based approach for estimating billboard dimensions from a single mobile photograph. The proposed system combines computer vision techniques to detect billboards, estimate width and height, and calculate surface area automatically. The prototype was evaluated on real-world billboard images and demonstrated confidence scores above 85% in successful detections.

Introduction

Manual billboard measurement is time-consuming and resource-intensive. This study investigates the use of AI and computer vision to automate billboard measurement using a smartphone image.

Problem Statement

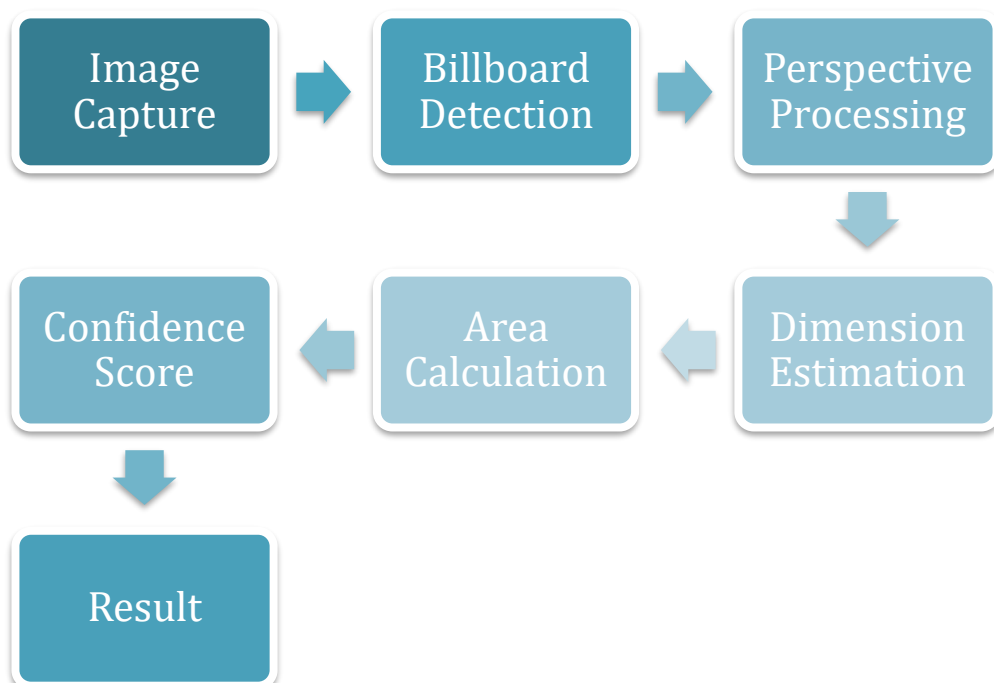
Develop an AI-assisted system capable of estimating billboard dimensions from a single image with practical accuracy for commercial use.

Objectives

- Detect billboards automatically.
- Estimate width, height and area.
- Generate confidence score.
- Evaluate real-world feasibility.

Methodology

Workflow:

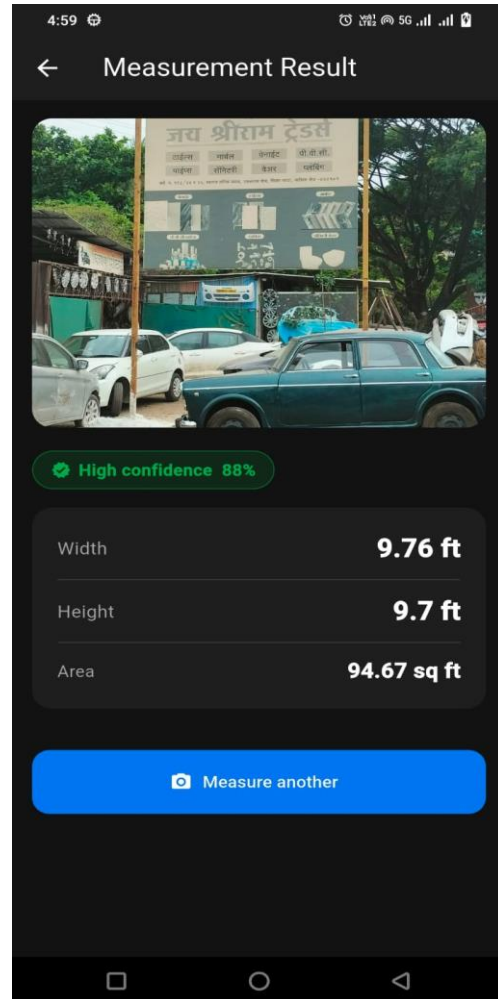


Experimental Results

The prototype successfully estimated billboard dimensions in real-world conditions. A representative test measured Width: 9.76 ft, Height: 9.70 ft, Area: 94.67 sq ft with an 88% confidence score.



Pic1: Clicked/Uploaded Image



Pic2: Result of size estimation

Applications

Outdoor advertising, smart cities, infrastructure monitoring, municipal auditing, digital asset management, and GIS-assisted planning.

Limitations

Performance may vary due to occlusion, extreme viewing angles, lighting conditions, and distance estimation errors.

Ethical & Governance Considerations

AI-generated measurements should be transparent, support human verification, and respect privacy during image collection.

Future Scope

Integration with AR, LiDAR, GIS, drone imagery, explainable AI, and automated compliance monitoring.

Conclusion

The study demonstrates that computer vision can automate billboard measurement using a single mobile image and supports future AI-enabled infrastructure assessment.